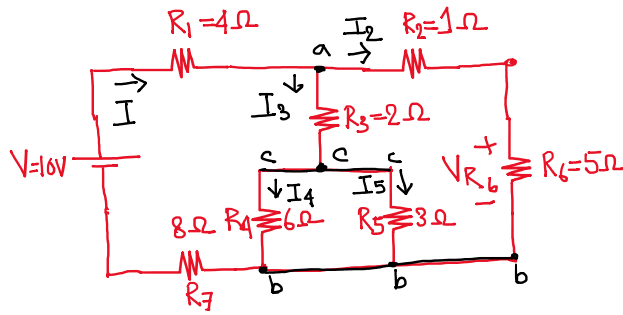
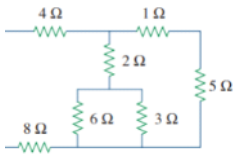
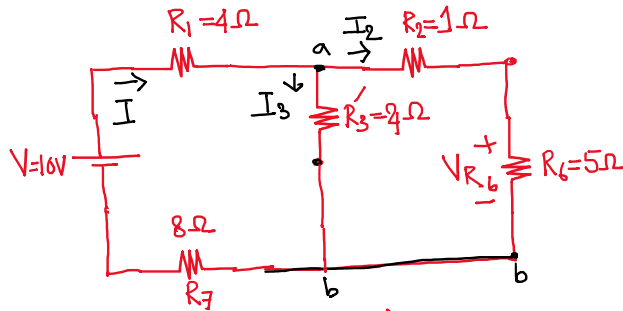
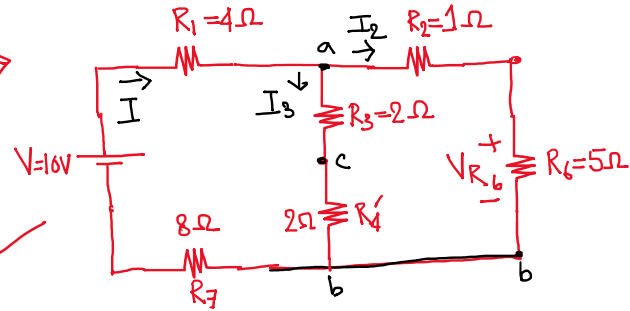




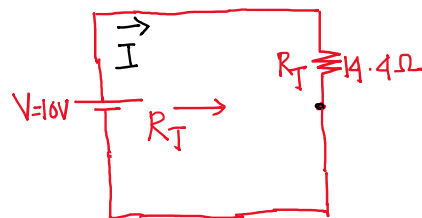
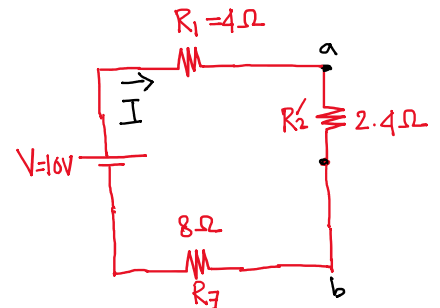
Between node b & c: $R_4 \parallel R_5 = R'_4$
 $R'_4 = 6 \parallel 3 = 2\Omega = \frac{3 \times 6}{9} \Omega$

$a \rightarrow b: R_3 + R'_4 = R'_3 = 2 + 2 = 4\Omega$



Between a & b: $R'_2 = R'_3 \parallel (R_2 + R_6)$
 $= 4 \parallel 6$
 $= \frac{24}{10} = 2.4\Omega$

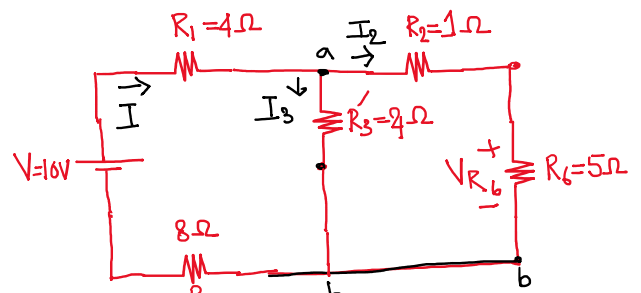
$R_T = R_1 + R'_2 + R_7 = 4 + 2.4 + 8 = 14.4\Omega$



$I = \frac{V}{R_T} = \frac{10}{14.4} = 0.694A$

Return

$I_2 = \frac{R'_3}{R'_3 + (R_2 + R_6)} I$
 $I_2 = \frac{4}{4 + 1 + 5} \times 0.694$
 $= \frac{4}{10} \times 0.694$



$$= \frac{4+1+5}{4} \times 0.694$$

$$= \frac{10}{10} \times 0.694$$

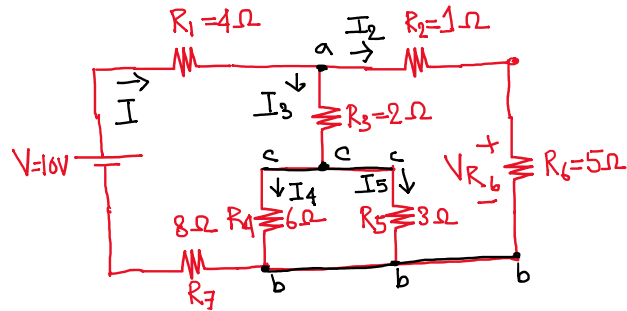
$$= 0.27 \text{ A}$$

$$I_3 = \frac{R_2 + R_6}{R_3 + R_2 + R_6} \times I = \frac{6}{10} \times 0.694$$

$$= \cancel{0.476} 0.416 \text{ A}$$

$$V_{R_6} = I_2 R_6 = 0.27 \times 5$$

$$= 1.385 \text{ V}$$



$$I_5 = \frac{R_4}{R_4 + R_5} \times I_3$$

$$= \frac{6}{9} \times 0.277$$

$$= 0.184 \text{ A}$$